ASSIGNMENT - 3

NUMERICAL TECHNIQUES IN HYDRODYNAMICS.

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OE23S018

Problem statement :-

Solve the poisson's equations using the Boundary condition and sources. Test the solution for the different values of A = 0.05, 0.01, 0.1.

Aim :-

To solve the poisson's equations and varying the values of A under the Jacobi Iteration, Gauss Siedel, Successive Under Relaxation methods.

Poissons equation :-

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = Q(x,y).$$

Boundary conditions :-

Case : 1

$T = 0$ at $y = 0$ and $y = 1.0$

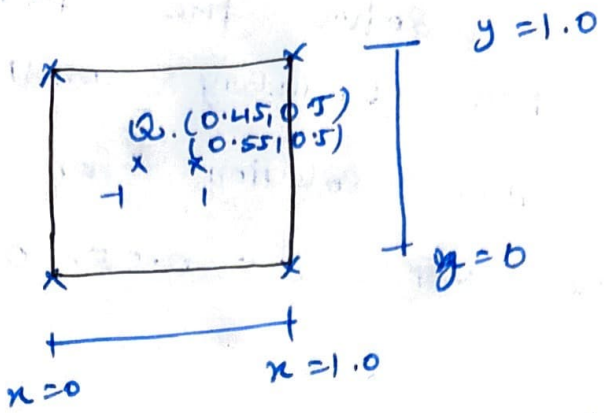
$T = 10^\circ$ at $x = 0$ and $x = 1.0$

Case : 2.

$T = 0$ at $y = 0$ and $y = 1.0$

$T = 0^\circ$ at $x = 0$ and $\frac{\partial T}{\partial x} = -1.0$ at $x = 1.0$.

Schematic 2D Heat solution :-



Finite difference method For solution :-

Governing (or) general poissons equations :-

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = Q(x,y).$$

using the central difference method

$$\frac{T_{i-1j} + T_{i+1j} - 2T_{ij}}{\Delta x^2} + \frac{T_{ij-1} + T_{ij+1} - 2T_{ij}}{\Delta y^2} = Q(x,y).$$

Jacobi Iteration method :-

by solving the above equation and the uniform mesh $\Delta x = \Delta y = \Delta$

$$T_{ij}^{n+1} = \frac{1}{4}\left[\left[T_{i+1j}^{n} + T_{i-1j}^{n} + T_{ij-1}^{n} + T_{ij+1}^{n}\right] - \Delta x^2 [Q_{ij}].\right]$$

Gauss Siedel Iteration method :-

$$T_{ij}^{n+1} = \frac{1}{4}\left[T_{i+1j}^{n+1} + T_{i+1j}^{n} + T_{ij-1}^{n+1} + T_{ij+1}^{n} - \Delta x^2 (Q_{ij}) \right]$$

Successive under Relaxation :-

$$T_{ij}^{n+1} = \frac{1}{4}\left[T_{ij}^{n+1} + T_{i+1j}^{n} + T_{ij-1}^{n+1} + T_{ij+1}^{n} - \Delta x^2 Q(x,y) \right]$$

under Relaxation.

$$T_{ij}^{n+1} = T_{ij}^{n} + \alpha \left[T_{ij}^{n+1} - T_{ij}^{n} \right].$$